Cheng Chen

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Education

Institute of Automation, Chinese Academy of Sciences Sept. 2027 (expected)
Incoming Ph.D. Student, Beijing, China. Supervisor: **Prof. Kaiqi Huang**.

Tongji University Sept. 2023 - Jun. 2027 (expected)
B.Eng. in Communication Engineering, Shanghai, China.
Average score: **92.02/100** GPA: **4.70** Major rank: **2/85**

Main interests

My current research interests center on finding simpler, more elegant ways to improve model capabilities. Areas of interest include, but are not limited to, multimodal large language models, video and image generation, and embodied intelligence.

Publications

* Equal contribution; † corresponding author.

1) **If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization.**
Cheng Chen*, Yi Xu*, Wenzhuo Lei.
ACM MM 2026, accepted. [PDF]
Separates supervision from visual and audio teachers for open-vocabulary audio-visual event localization.

2) **OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment.**
Yi Xu*, **Cheng Chen***, Mufan Cao.
IJCAI 2026, accepted. [PDF]
Distills complementary clinical knowledge from heterogeneous teachers into a lightweight retinal-image classifier.

3) **Enabling Visual Reasoning in Text-Only LLMs via Symbolic Scene Descriptions.**
Wenzhuo Lei, Mufan Cao, Chaorong Ye, Yi Xu, **Cheng Chen**†.
IJCNN 2026, accepted. [PDF]
Links object detection to text-only language models using compact symbolic scene descriptions.

Preprints

4) From Scores to Checks: A Human-Annotated Rubric Benchmark for AI-Generated Image Faithfulness. *Preprint, 2026.*
Uses human-annotated rubric checks to evaluate whether generated images satisfy individual prompt requirements.

5) What Can Activation-Patching Evidence Establish? *Preprint, 2026.*
Examines the scope and limitations of the functional claims that activation-patching evidence supports.

6) Follow Me If You Can: Auditing Understanding-to-Generation Transfer. *Preprint, 2026.*
Audits transfer from visual-understanding training to image generation through controlled comparisons.

7) Allocation Entropy for Single-Pass Risk Triage in Neural Operators. *Preprint, 2026.*
Studies single-pass screening of unreliable neural-operator predictions without altering the predicted physical field.

Research internships

Research Intern
Hong Kong University of Science and Technology (Guangzhou) Sept. 2026 - Present
Zhejiang University Jun. 2026 - Sept. 2026
Tongji University Oct. 2025 - Jun. 2026

Academic service

Reviewer: ICLR, PRCV, ICME, IJCNN, AAAI.

Research experience

If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization *ACM MM 2026; co-first author*

- Routed strong visual supervision to the localization path and weaker audio supervision to an orthogonal auxiliary path; used text prototypes to support unseen event categories.
- Reached **0.816 segment average precision** on OV-AVEBench. F1@0.5 improved over the official fine-tuning baseline by **2.7 points** overall and **3.4 points** on unseen categories.

OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment *IJCAI 2026; co-first author*

- Combined full supervision from a stronger teacher with representation-level transfer from a weaker teacher; orthogonal student projections preserved complementary clinical information.
- Used a combined pool of **132,049** retinal images. A **5.4M**-parameter student reached **0.885 QWK** (quadratic weighted kappa) on EyePACS and improved zero-shot Messidor-2 QWK from **0.507 to 0.728**.

Enabling Visual Reasoning in Text-Only LLMs via Symbolic Scene Descriptions *IJCNN 2026; corresponding author*

- Proposed the initial idea and worked with a junior student, who led implementation and experiments, to connect an object detector with a text-only language model through compact scene reports.
- The pipeline reached **42.3%** visual question answering accuracy on a stratified **5,000-image VQAv2 subset**, versus **38.2%** for scene-graph serialization, using **1/2.5 as many prompt tokens as that baseline**.

From Scores to Checks: A Human-Annotated Rubric Benchmark for AI-Generated Image Faithfulness *2026*

- Studied whether generated images satisfy individual requirements in text prompts, using human annotations to separate different types of faithfulness errors.
- Compared lightweight evaluators under matched conditions; examined confidence calibration, supporting image regions, and transfer across generators and datasets.

What Can Activation-Patching Evidence Establish? *2026*

- Studied which functional claims observations from neural-network interventions can support, including cases where matching observed outputs leaves other behavior unresolved.
- Combined mathematical analysis with controlled language-model tests to examine additional measurements and failures on previously unmeasured responses.

Follow Me If You Can: Auditing Understanding-to-Generation Transfer *2026*

- Audited whether training for visual understanding changes image generation in unified multimodal models, focusing on counting and spatial relations.
- Used paired model states and controlled comparisons to distinguish gains in answering questions, effects on generation, and transfer of the requested visual attribute.

Allocation Entropy for Single-Pass Risk Triage in Neural Operators *2026*

- Studied risk indicators available within one prediction pass of a neural operator, preserving the predicted physical field and avoiding a separate learned risk head.
- Tested how the scoring representation affects error ranking and acceptance decisions across physical tasks; compared screening quality and computational cost with alternative indicators.

Awards and honors

- National First Prize, China Robot and Artificial Intelligence Competition (2026); Team Captain.
- National Second Prize, Global Campus AI Algorithm Elite Competition (2025); Team Captain.
- National Third Prize, Embedded Chip and System Design Competition, FPGA Track (2025); Team Captain.
- Provincial Third Prize, National Mathematical Modeling Competition (2025); Team Captain.
- Provincial Third Prize, National Statistical Modeling Competition (2025); Team Captain.
- Provincial Second Prize, National College Student Mathematics Competition (2024–2025).
- Tongji University Outstanding Student Scholarship, Second Class (2023–2024 and 2024–2025).

Technical skills

Programming: Python, C/C++, Verilog. **Machine learning:** PyTorch.
Tools: Git, Linux, LaTeX, OpenAI Codex, Claude Code.
Foundations: Machine learning, algorithms, linear algebra, probability, optimization, and signal processing.